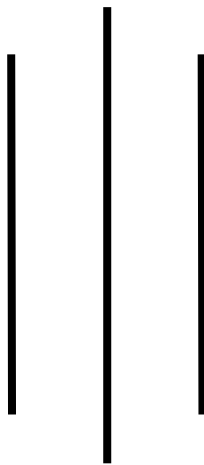


TRIBHUVAN UNIVERSITY

PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR



NET CENTRIC COMPUTING (BIT 351)

SUBMITTED BY:

NAME: SURESH DAHAL

ROLL: 23

CLASS: BIT-III/II

DATE: 2083/05/08

SUBMITTED TO:

SUBASH SIWA

CHECKED BY

UNIT-2

1. Frameworks & CLI Initialization

Step 1: Create a new Web API project

```
D:\Online Class\suresh>dotnet new webapi -n MyWebApiProject
The template "ASP.NET Core Web API" was created successfully.

Processing post-creation actions...
Restoring D:\Online Class\suresh\MyWebApiProject\MyWebApiProject.csproj:
Restore succeeded with 1 warning(s) in 2.2s
    D:\Online Class\suresh\MyWebApiProject\MyWebApiProject.csproj : warning NU1903:
    Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://
    /github.com/advisories/GHSA-v5pm-xwqc-g5wc
Restore succeeded.
```

Step 2: Navigate into the project directory

```
D:\Online Class\suresh>cd MyWebApiProject
D:\Online Class\suresh\MyWebApiProject>
```

Step 3: Compile the application

```
D:\Online Class\suresh\MyWebApiProject>dotnet build
Restore succeeded with 1 warning(s) in 0.6s
    D:\Online Class\suresh\MyWebApiProject\MyWebApiProject.csproj : warning NU1903:
    Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://
    /github.com/advisories/GHSA-v5pm-xwqc-g5wc
    MyWebApiProject net10.0 succeeded with 1 warning(s) (4.4s) → bin\Debug\net10.0\My
    WebApiProject.dll
    D:\Online Class\suresh\MyWebApiProject\MyWebApiProject.csproj : warning NU1903:
    Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://
    /github.com/advisories/GHSA-v5pm-xwqc-g5wc
Build succeeded with 2 warning(s) in 5.9s
```

Step 4: Execute / Run the application

```
D:\Online Class\suresh\MyWebApiProject>dotnet run
Using launch settings from D:\Online Class\suresh\MyWebApiProject\Properties\launch
Settings.json...
Building...
Restore succeeded with 1 warning(s) in 0.5s
    D:\Online Class\suresh\MyWebApiProject\MyWebApiProject.csproj : warning NU1903:
    Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://
    /github.com/advisories/GHSA-v5pm-xwqc-g5wc
    MyWebApiProject net10.0 succeeded with 1 warning(s) (0.5s) → bin\Debug\net10.0\My
    WebApiProject.dll
    D:\Online Class\suresh\MyWebApiProject\MyWebApiProject.csproj : warning NU1903:
    Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://
    /github.com/advisories/GHSA-v5pm-xwqc-g5wc
info: Microsoft.Hosting.Lifetime[14]
    Now listening on: http://localhost:5175
info: Microsoft.Hosting.Lifetime[0]
    Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
    Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
    Content root path: D:\Online Class\suresh\MyWebApiProject
```

2. Compilation, MSIL, and CLR

Step 1: Create a basic C# Console Application

```
D:\Online Class\suresh>dotnet new console -n MyConsoleApp
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring D:\Online Class\suresh\MyConsoleApp\MyConsoleApp.csproj:
Restore succeeded.
```

```
D:\Online Class\suresh>cd MyConsoleApp
D:\Online Class\suresh\MyConsoleApp>dotnet build
```

Step 2: Compile the application

```
D:\Online Class\suresh\MyConsoleApp>dotnet build
Restore complete (0.4s)
  MyConsoleApp net10.0 succeeded (0.5s) → bin\Debug\net10.0\MyConsoleApp.dll
Build succeeded in 1.6s
```

Step 3: Locate the MSIL Output File

```
D:\Online Class\suresh\MyConsoleApp>cd bin/Debug/net10.0/
D:\Online Class\suresh\MyConsoleApp\bin\Debug\net10.0>dir
Volume in drive D is Coding
Volume Serial Number is F222-F771

Directory of D:\Online Class\suresh\MyConsoleApp\bin\Debug\net10.0

08/22/2026  02:17 PM    <DIR>          .
08/22/2026  02:17 PM    <DIR>          ..
08/22/2026  02:17 PM                430 MyConsoleApp.deps.json
08/22/2026  02:17 PM            4,608 MyConsoleApp.dll
08/22/2026  02:17 PM           162,304 MyConsoleApp.exe
08/22/2026  02:17 PM           10,812 MyConsoleApp.pdb
08/22/2026  02:17 PM                270 MyConsoleApp.runtimeconfig.json
               5 File(s)            178,424 bytes
               2 Dir(s)  57,048,563,712 bytes free
```

3. .NET Architecture & Deployment

Step 1: Create a basic ASP.NET Core application

```
D:\Online Class\suresh>dotnet new web -n DeployApp
The template "ASP.NET Core Empty" was created successfully.

Processing post-creation actions...
Restoring D:\Online Class\suresh\DeployApp\DeployApp.csproj:
Restore succeeded.
```

```
D:\Online Class\suresh>cd DeployApp
```

```
D:\Online Class\suresh\DeployApp>
```

Step 2: Publish the application for production (Release mode)

```
D:\Online Class\suresh\DeployApp>dotnet publish -c Release -o ./publish
Restore complete (0.5s)
  DeployApp net10.0 succeeded (0.9s) → publish\

Build succeeded in 2.2s
```

Step 3: Release output folder

```
D:\Online Class\suresh\DeployApp\bin\Release\net10.0>dir
Volume in drive D is Coding
Volume Serial Number is F222-F771

Directory of D:\Online Class\suresh\DeployApp\bin\Release\net10.0

08/22/2026  02:22 PM    <DIR>          .
08/22/2026  02:22 PM    <DIR>          ..
08/22/2026  02:21 PM             127 appsettings.Development.json
08/22/2026  02:21 PM             151 appsettings.json
08/22/2026  02:22 PM             421 DeployApp.deps.json
08/22/2026  02:22 PM             5,120 DeployApp.dll
08/22/2026  02:22 PM          162,304 DeployApp.exe
08/22/2026  02:22 PM          21,160 DeployApp.pdb
08/22/2026  02:22 PM             491 DeployApp.runtimeconfig.json
08/22/2026  02:22 PM             51 DeployApp.staticwebassets.endpoints.json
               8 File(s)          189,825 bytes
               2 Dir(s)  57,048,207,360 bytes free
```

Publish

Where are you publishing today?

Target



Azure

Host your application to the Microsoft cloud



Docker Container Registry

Publish your application to any supported Container Registry that works with Docker images



Folder

Publish your application to a local folder or file share



FTP/FTPS Server

Publish your application to an FTP/FTPS server



Web Server (IIS)

Publish your application to IIS using Web Deploy or Web Deploy Package



Import Profile

Import your publish settings to deploy your app

4. Web Forms & Automated Testing

Step 1: Create a solution and add a Class Library

```
D:\Online Class\suresh>dotnet new sln -n MyTestSolution
The template "Solution File" was created successfully.
```

```
D:\Online Class\suresh>dotnet new classlib -n MyClassLib
The template "Class Library" was created successfully.

Processing post-creation actions...
Restoring D:\Online Class\suresh\MyClassLib\MyClassLib.csproj:
Restore succeeded.
```

```
D:\Online Class\suresh>dotnet sln add MyClassLib/MyClassLib.csproj
Project 'MyClassLib\MyClassLib.csproj' added to the solution.
```

Step 2: Create an xUnit Test project and add reference

```
D:\Online Class\suresh>dotnet new xunit -n MyTestProject
The template "xUnit Test Project" was created successfully.

Processing post-creation actions...
Restoring D:\Online Class\suresh\MyTestProject\MyTestProject.csproj:
Restore succeeded.
```

```
D:\Online Class\suresh>dotnet sln add MyTestProject/MyTestProject.csproj
Project 'MyTestProject\MyTestProject.csproj' added to the solution.
```

```
D:\Online Class\suresh>dotnet add MyTestProject/MyTestProject.csproj reference MyCl
assLib/MyClassLib.csproj
Reference '..\MyClassLib\MyClassLib.csproj' added to the project.
```

Step 3: Write a basic assertion test (in MyTestProject/UnitTest1.cs)

```
D:\Online Class\suresh>copy con UnitTest1.cs
using Xunit;

public class UnitTest1
{
    [Fact]
    public void Test1()
    {
        int a = 5;
        int b = 5;
        Assert.Equal(10, a + b);
    }
}
^Z          1 file(s) copied.
```

Step 4: Execute the test suite

```
D:\Online Class\suresh>dotnet test
Restore complete (2.1s)
  MyClassLib net10.0 succeeded (3.7s) → MyClassLib\bin\Debug\net10.0\MyClassLib.dll
  MyTestProject net10.0 succeeded (2.3s) → MyTestProject\bin\Debug\net10.0\MyTestPr
object.dll
[xUnit.net 00:00:00.00] xUnit.net VSTest Adapter v3.1.4+50e68bbb8b (64-bit .NET 10.
0.11)
[xUnit.net 00:00:00.53]   Discovering: MyTestProject
[xUnit.net 00:00:00.59]   Discovered:   MyTestProject
[xUnit.net 00:00:00.61]   Starting:     MyTestProject
[xUnit.net 00:00:00.67]   Finished:     MyTestProject
  MyTestProject test net10.0 succeeded (3.9s)

Test summary: total: 1, failed: 0, succeeded: 1, skipped: 0, duration: 3.8s
Build succeeded in 12.3s
```

UNIT-3

1. HTTP Communication and ASP.NET Core Web API Development

Step 1: Create a Web API Project

```
D:\Online Class\suresh>dotnet new webapi -n HttpApiApp
The template "ASP.NET Core Web API" was created successfully.

Processing post-creation actions...
Restoring D:\Online Class\suresh\HttpApiApp\HttpApiApp.csproj:
Restore succeeded.
```

```
D:\Online Class\suresh>cd HttpApiApp
D:\Online Class\suresh\HttpApiApp>
```

Step 2: Create Model Class (Models/Student.cs)

```
D:\Online Class\suresh\HttpApiApp>mkdir models && cd models
D:\Online Class\suresh\HttpApiApp\models>copy con Student.cs
namespace HttpApiApp.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Department { get; set; }
    }
}
^Z          1 file(s) copied.
```

Step 3: Create API Controller (Controllers/StudentsController.cs)

```
D:\Online Class\suresh\HttpApiApp\controllers>copy con StudentController.cs
using Microsoft.AspNetCore.Mvc;
using HttpApiApp.Models;

[Route("api/[controller]")]
[ApiController]
public class StudentsController : ControllerBase
{
    private static List<Student> students = new()
    {
        new Student { Id = 1, Name = "Suresh", Department = "CS" },
        new Student { Id = 2, Name = "Luna", Department = "IT" }
    };
}
```

Swagger Documentation

HttpApiApp

1.0

OAS 3.0

<https://localhost:44380/swagger/v1/swagger.json>

Students

GET /api/Students

POST /api/Students

GET /api/Students/{id}

PUT /api/Students/{id}

DELETE /api/Students/{id}

Request URL

<https://localhost:44380/api/Students>

Server response

Code	Details
------	---------

200	
-----	--

Response body

```
[
  {
    "id": 1,
    "name": "Suresh",
    "department": "CS"
  },
  {
    "id": 2,
    "name": "Luna",
    "department": "IT"
  }
]
```

Response headers

```
content-type: application/json; charset=utf-8
date: Sat, 22 Aug 2026 09:54:23 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

2. ASP.NET Core Architecture, MVC Pattern, Projects, and Application Development

Create an MVC Project

```
D:\Online Class\suresh>dotnet new mvc -n MvcApp
The template "ASP.NET Core Web App (Model-View-Controller)" was created successfully.
This template contains technologies from parties other than Microsoft, see https://aka.ms/aspnetcore/10.0-third-party-notices for details.

Processing post-creation actions...
Restoring D:\Online Class\suresh\MvcApp\MvcApp.csproj:
Restore succeeded.
```

```
D:\Online Class\suresh>cd MvcApp
```

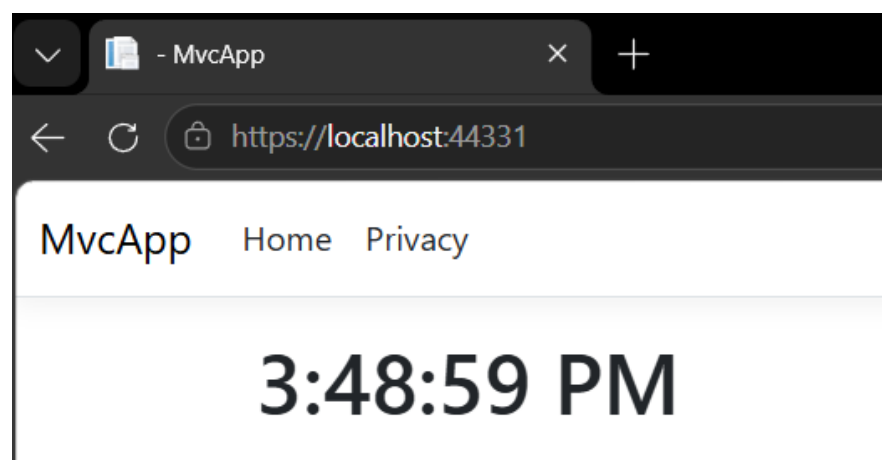
```
D:\Online Class\suresh\MvcApp>
```

Step 2: Register Custom Service & middleware

```
Directory of D:\Online Class\suresh\MvcApp

08/22/2026  03:45 PM    <DIR>          .
08/22/2026  03:40 PM    <DIR>          ..
08/22/2026  03:40 PM                127 appsettings.Development.json
08/22/2026  03:40 PM                151 appsettings.json
08/22/2026  03:42 PM    <DIR>          bin
08/22/2026  03:40 PM    <DIR>          Controllers
08/22/2026  03:45 PM                70 Middleware.cs
08/22/2026  03:40 PM    <DIR>          Models
08/22/2026  03:40 PM                220 MvcApp.csproj
08/22/2026  03:42 PM                238 MvcApp.csproj.user
08/22/2026  03:42 PM    <DIR>          obj
08/22/2026  03:44 PM                761 Program.cs
08/22/2026  03:40 PM    <DIR>          Properties
08/22/2026  03:43 PM                67 Service.cs
08/22/2026  03:40 PM    <DIR>          Views
08/22/2026  03:40 PM    <DIR>          wwwroot
                                7 File(s)      1,634 bytes
```

Output



UNIT-4

1. Developing an ASP.NET Core MVC Application with Controllers, Actions, Views, and Model Validation.

PROGRAM:

Index.cshtml

```
@model IEnumerable<MvcValidationApp.Models.Employee>

@{
    ViewData["Title"] = "Employees";
}

<div class="container mt-4">

    <div class="d-flex justify-content-between align-items-center mb-3">
        <h1>Employee Index</h1>

        <a asp-controller="Employee"
            asp-action="Create"
            class="btn btn-success">
            Create Record
        </a>
    </div>

    <h2>All Records</h2>

    @if (!Model.Any())
    {
        <div class="alert alert-info">
            No employees found.
        </div>
    }
    else
    {
        <table class="table table-bordered table-striped">
            <thead class="table-dark">
                <tr>
                    <th>ID</th>
                    <th>Name</th>
                    <th style="width: 250px;">Actions</th>
                </tr>
            </thead>

            <tbody>
                @foreach (var emp in Model)
                {
                    <tr>
                        <td>@emp.Id</td>
                        <td>@emp.Name</td>

                        <td>
                            <a asp-action="Details"
                                asp-route-id="@emp.Id"
                                class="btn btn-sm btn-info">
                                Details
                            </a>
                        </td>
                    </tr>
                }
            </tbody>
        </table>
    }
}
```

```

        <a asp-action="Edit"
            asp-route-id="@emp.Id"
            class="btn btn-sm btn-primary">
            Edit
        </a>

        <form asp-action="Delete"
            asp-route-id="@emp.Id"
            method="post"
            style="display:inline;"
            onsubmit="return confirm('Are you sure you
want to delete this employee?');">

            @Html.AntiForgeryToken()

            <button type="submit"
                class="btn btn-sm btn-danger">
                Delete
            </button>
        </form>
    </td>
</tr>
}
</tbody>
</table>
}
</div>

```

Create.cshtml

```

@model MvcValidationApp.Models.Employee

@{
    ViewData["Title"] = "Create Employee";
}

<div class="container mt-4">
    <h1>Create Employee</h1>

    <form asp-action="Create" method="post">
        @Html.AntiForgeryToken()

        <div asp-validation-summary="ModelOnly"
            class="text-danger mb-3">
        </div>

        <div class="mb-3">
            <label asp-for="Id" class="form-label"></label>

            <input asp-for="Id"
                class="form-control" />

            <span asp-validation-for="Id"
                class="text-danger">

```

```

        </span>
    </div>

    <div class="mb-3">
        <label asp-for="Name" class="form-label"></label>

        <input asp-for="Name"
            class="form-control" />

        <span asp-validation-for="Name"
            class="text-danger">
        </span>
    </div>

    <button type="submit"
        class="btn btn-success">
        Save
    </button>

    <a asp-action="Index"
        class="btn btn-secondary">
        Back
    </a>

</form>

</div>

```

Edit.cshtml

```

@model MvcValidationApp.Models.Employee

@{
    ViewData["Title"] = "Edit Employee";
}

<div class="container mt-4">

    <h1>Edit Employee</h1>

    <form asp-action="Edit"
        asp-route-id="@Model.Id"
        method="post">

        @Html.AntiForgeryToken()

        <div asp-validation-summary="ModelOnly"
            class="text-danger mb-3">
        </div>

        <div class="mb-3">
            <label asp-for="Id" class="form-label"></label>

            <input asp-for="Id"
                class="form-control"
                readonly />

            <span asp-validation-for="Id"
                class="text-danger">

```

```

        </span>
    </div>

    <div class="mb-3">
        <label asp-for="Name" class="form-label"></label>

        <input asp-for="Name"
            class="form-control" />

        <span asp-validation-for="Name"
            class="text-danger">
        </span>
    </div>

    <button type="submit"
        class="btn btn-primary">
        Update
    </button>

    <a asp-action="Index"
        class="btn btn-secondary">
        Cancel
    </a>

</form>

</div>

```

EmployeeController.cs

```

using Microsoft.AspNetCore.Mvc;
using MvcValidationApp.Models;
using System.Collections.Generic;
using System.Linq;

public class EmployeeController : Controller
{
    private static List<Employee> employees = new List<Employee>();

    [HttpGet]
    public IActionResult Index()
    {
        return View(employees);
    }

    [HttpGet]
    public IActionResult Details(int id)
    {
        var employee = employees.FirstOrDefault(e => e.Id == id);

        if (employee == null)
        {
            return NotFound();
        }

        return View(employee);
    }

    [HttpGet]
    public IActionResult Create()

```

```

    {
        return View();
    }

    [HttpPost]
    [ValidateAntiForgeryToken]
    public IActionResult Create(Employee emp)
    {
        if (!ModelState.IsValid)
        {
            return View(emp);
        }

        employees.Add(emp);

        return RedirectToAction(nameof(Index));
    }

    [HttpGet]
    public IActionResult Edit(int id)
    {
        var employee = employees.FirstOrDefault(e => e.Id == id);

        if (employee == null)
        {
            return NotFound();
        }

        return View(employee);
    }

    [HttpPost]
    [ValidateAntiForgeryToken]
    public IActionResult Edit(int id, Employee emp)
    {
        if (id != emp.Id)
        {
            return BadRequest();
        }

        if (!ModelState.IsValid)
        {
            return View(emp);
        }

        var employee = employees.FirstOrDefault(e => e.Id == id);

        if (employee == null)
        {
            return NotFound();
        }

        employee.Name = emp.Name;

        return RedirectToAction(nameof(Index));
    }

    [HttpPost]
    [ValidateAntiForgeryToken]
    public IActionResult Delete(int id)

```

```

    {
        var employee = employees.FirstOrDefault(e => e.Id == id);

        if (employee == null)
        {
            return NotFound();
        }

        employees.Remove(employee);

        return RedirectToAction(nameof(Index));
    }

    [HttpGet]
    public ContentResult GetText()
    {
        return Content("This is plain content text.");
    }
}

```

Employee.cs

```

using System.ComponentModel.DataAnnotations;

namespace MvcValidationApp.Models
{
    public class Employee
    {
        public int Id { get; set; }

        [Required(ErrorMessage = "Name is required")]
        [StringLength(50, MinimumLength = 3)]
        public string Name { get; set; }
    }
}

```

OUTPUT:

Step 1: Create an MVC Project

```

D:\Online Class\suresh>dotnet new mvc -n MvcValidationApp
The template "ASP.NET Core Web App (Model-View-Controller)" was created successfully.
This template contains technologies from parties other than Microsoft, see https://aka.ms/aspnetcore/10.0-third-party-notices for details.

Processing post-creation actions...
Restoring D:\Online Class\suresh\MvcValidationApp\MvcValidationApp.csproj:
Restore succeeded.

```

```

D:\Online Class\suresh>cd MvcValidationApp
D:\Online Class\suresh\MvcValidationApp>

```

Step 2: Create a Validated Model

```
D:\Online Class\suresh\MvcValidationApp>cd models && copy con Employee.cs
using System.ComponentModel.DataAnnotations;

namespace MvcValidationApp.Models
{
    public class Employee
    {
        public int Id { get; set; }

        [Required(ErrorMessage = "Name is required")]
        [StringLength(50, MinimumLength = 3)]
        public string Name { get; set; }

        [Required]
        [EmailAddress(ErrorMessage = "Invalid Email Address")]
        public string Email { get; set; }
    }
}
^Z          1 file(s) copied.
```

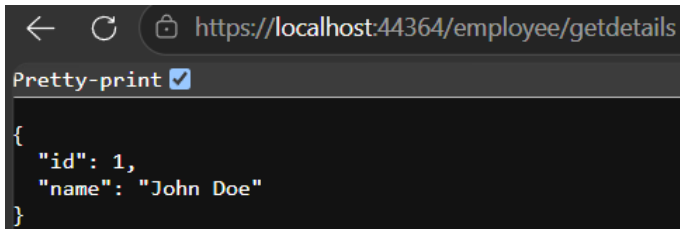
Step 3: Implement Controller Demonstrating Action Results & Validation

```
D:\Online Class\suresh\MvcValidationApp\Controllers>copy con EmployeeController.cs
using Microsoft.AspNetCore.Mvc;
using MvcValidationApp.Models;

public class EmployeeController : Controller
{
    // GET: Employee/Details/5
    public IActionResult Details(int? id)
```

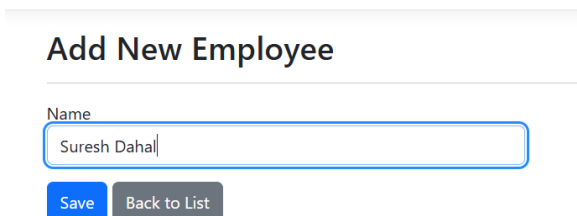
Output:

Json Response



```
<  ↻  https://localhost:44364/employee/getdetails
Pretty-print ☒
{
  "id": 1,
  "name": "John Doe"
}
```

Add new record



Add New Employee

Name

Suresh Dahal

Save Back to List

View records

Employee Index

[Create Record](#)

All Records:

ID	Name	Actions
0	Suresh Dahal	Edit Delete

Delete record

[Home](#) [Privacy](#)

Employee Index

Records:

Name	Actions
Another record	Edit Delete

View details

Employee Details

Id	1
Name	Suresh Dahal
Edit Back to List	

Update record

Edit Employee

Id	1
Name	Suresh Prasad Dahal
Update Cancel	

2. Developing ASP.NET Core Web API with JSON, Dependency Injection, and IoC Container

Step 1: Create a Web API Project

```
D:\Online Class\suresh\MvcValidationApp>cd ..  
  
D:\Online Class\suresh>dotnet new webapi -n ApiDiApp  
The template "ASP.NET Core Web API" was created successfully.  
  
Processing post-creation actions...  
Restoring D:\Online Class\suresh\ApiDiApp\ApiDiApp.csproj:  
Restore succeeded.
```

```
D:\Online Class\suresh>cd ApiDiApp  
  
D:\Online Class\suresh\ApiDiApp>
```

Step 2: Create a Model Class

```
D:\Online Class\suresh\ApiDiApp\Models>copy con Product.cs  
namespace ApiDiApp.Models  
{  
    public class Product  
    {  
    }  
}
```

Step 3: Create Service Interface and Implementation

```
Directory of D:\Online Class\suresh\ApiDiApp\Services  
  
08/22/2026  05:15 PM    <DIR>          .  
08/22/2026  05:14 PM    <DIR>          ..  
08/22/2026  05:15 PM                86 ProductService.cs  
                                1 File(s)        86 bytes
```

Step 4: Register Services in IoC Container

```
var app = builder.Build();  
  
builder.Services.AddTransient<IProductService, ProductService>();
```

Step 5: Create Product API Controller with DI and view with Swagger UI

- Controllers
 - ProductController.cs
- Models
 - Product.cs
- Services
 - ProductService.cs

Request URL

https://localhost:44336/api/Products

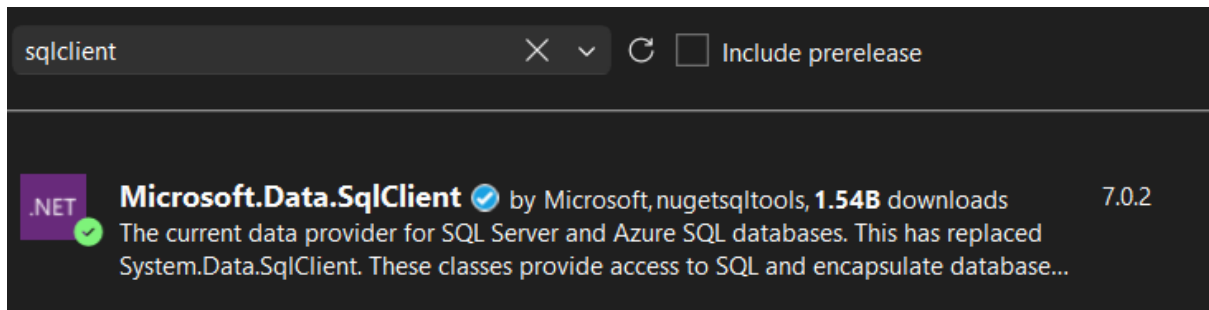
Server response

Code	Details
200	Response body <pre>[{ "productid": 1, "name": "Laptop", "price": 50000, "category": "Electronics" }]</pre>

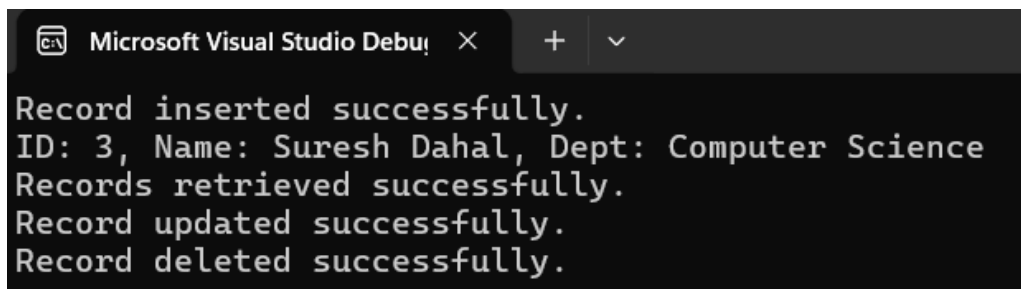
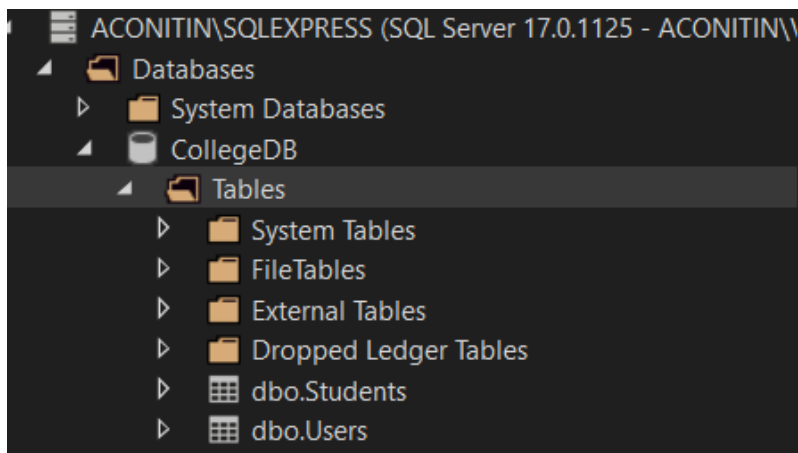
UNIT-5

1. Database Operations Using ADO.NET Classes.

Add package

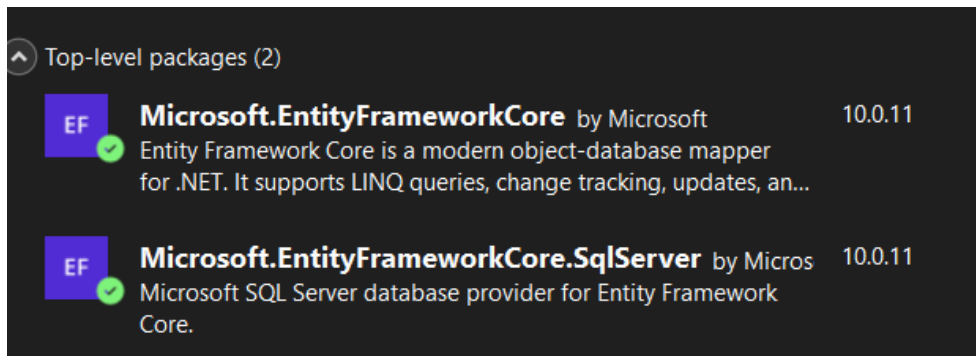


Create Database

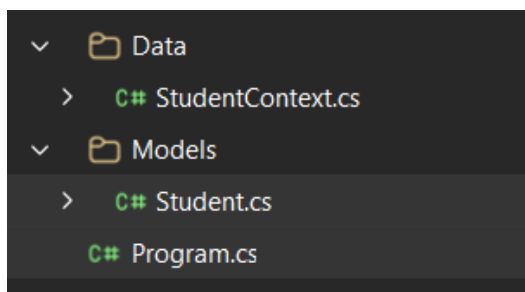


2. Database Application Development Using Entity Framework Core.

Step 1: Install NuGet Packages



Step 2: Create Model Class and Data Context Class



Step 4: Implement CRUD Operations

